Experiment No : 4

Title: .Write a program in C++ to perform following operations on complex number Add,Subtract, Multiply, Divide, Complex conjugate. Design the class for complexnumber representation and the operations to be performed.

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Program:

```
#include <iostream>
#include <math.h>
using namespace std;
class complex
{
  private:
     int real , imag ;

  public:
    void display();
    void read();
    complex add(complex);
    complex subtract(complex);
    complex multiply(complex);
};

void complex::read()
{
  cout << "\nEnter the real part:  ";
  cin >> real ;
  cout << "Enter the imaginary part:  ";
  cin >> imag ;
}

void complex::display()
{
    if (imag >= 0)
```

```
    cout << real << " + " << imag << "i" ;
  else
    cout << real << " - " << imag << "i" ;
}

complex complex::add(complex c1)
{
  complex sum ;
  sum.real = real + c1.real;
  sum.imag = imag + c1.imag;
  return sum;
}

complex complex::subtract(complex c1)
{
  complex differnce ;
  differnce.real = real - c1.real;
  differnce.imag = imag - c1.imag;
  return differnce;
}

complex complex::multiply(complex c1)
{
  complex product;
  product.real = (real* c1.real) - (imag* c1.imag);
  product.imag = (real* c1.imag) + (imag* c1.real);
  return product;
}

int main()
{
  complex c1 , c2 , c3 ;
  cout << "Enter first complex number " ;
  c1.read();
  cout << "\nEnter second complex number : " ;
  c2.read();
```

```
    cout << "\nFirst complex number : " ;
    c1.display();
    cout << "\nSecond complex number : " ;
    c2.display();

    c3 = c1.add(c2);
    cout << "\nAddition : " ;
    c3.display();

    c3 = c1.subtract(c2);
    cout << "\nSubtraction : " ;
    c3.display();

    c3 = c1.multiply(c2);
    cout << "\nMultiplication : " ;
    c3.display();
}
```

Output:

Output

```
Enter first complex number
Enter the real part:  5
Enter the imaginary part:  3

Enter second complex number :
Enter the real part:  4
Enter the imaginary part:  2

First complex number : 5 + 3i
Second complex number : 4 + 2i
Addition : 9 + 5i
Subtraction : 1 + 1i
Multiplication : 14 + 22i

--- Code Execution Successful ---
```